TECHNICAL APPROACH
Retail Behavioural Analytics

Prepared by: Twisha Khurana
Computer Vision / AI Engineer

Technical Methodology, Experiments, Failure Analysis and Results

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Technical Approach — Retail Behavioural Analytics

Setup, dependencies, commands, repository layout and output locations are in `README.md`. This document covers the engineering reasoning: how ambiguous behavioural requirements were turned into measurable events, what failed along the way, and where the evidence for the results ends.

> The main challenge was not detecting people. It was deciding what the behavioural terms in the brief meant in the footage, and then turning those definitions into rules that stayed consistent over time.

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1. Problem and Objective

Three behavioural analytics were required from two fixed-camera CCTV clips of a mall footwear store (1280×720, 30 fps, ~7.5 minutes each):

*1. **Store interest and conversion** (`entrance.mp4`) — how many passers-by show interest in the store, and whether they enter or continue past.*
*2. **Per-shelf customer interest** (`interior.mp4`) — discrete interest events per designated shelf.*
*3. **Staff–customer interaction** (`entrance.mp4`) — average interaction sessions per staff member.*

The detection and tracking components were pretrained models. Most of the engineering work was in translating the brief's terms — interest, engagement and interaction — into measurable rules. The brief defines the outcomes, but leaves the actual interpretation to the candidate.

Three questions came up repeatedly while designing the pipeline:

- **Who is a candidate?** For Task 1 the brief describes passers-by. Someone already shopping inside the store is not a passer-by, but a tracker does not know that.
- **When does one event become two?** A shopper who reaches into a shelf, withdraws to examine a shoe, and reaches back has performed one browse. The brief explicitly warns against counting that repeatedly.
- **What happens when the evidence is absent?** An occluded wrist is not evidence of not engaging. Treating missing observation as a negative fabricates events; treating it as a positive fabricates engagement.

One important constraint was that the footage has no labelled behavioural ground truth. I therefore based thresholds on the observed data, sensitivity checks, and properties of the scene rather than on accuracy or validation scores.

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2. Overall Methodology

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Detection  →  Tracking  →  Track-level observations  →  Temporal reasoning
                                                                ↓
                     Business metrics  ←  Behavioural events
```

Each stage answers a different question: where is the person, is this the same track as before, what has the track done over time, when does that behaviour become an event, and how should the event be counted? Keeping these parts separate also made it easier to trace a bad result back to the assumption that caused it.

This separation helped during development. I could change the tracker without rewriting the event logic because the task modules only use the resulting track observations. The same temporal logic is also reused for shelf engagement and staff-customer interaction: both need a minimum duration, tolerance for short gaps, protection against double counting, and a new event after a genuine return.

For this assessment, detection and tracking are run once per video and the resulting observations are cached. This became useful during debugging: because tracker IDs depend on the preceding frames, rendering a 45-second diagnostic clip starting at 300 seconds originally took about six minutes. Caching meant I could change event rules and re-run the analysis without repeating the full tracking pass.

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3. Task 1 — Store Interest and Conversion

Operationalising "interest"

The brief gives several possible signs of interest: looking or turning toward the storefront, slowing down, and approaching it. I treated these as separate cues and checked how useful each was in the actual footage:

- slowing — was clearly walking, then clearly slowed while outside. A clear slowdown in front of the shop is a useful browse signal. Requiring both ends of the transition prevents flagging someone who happened to be stationary throughout for unrelated reasons.
- **approaching** — closed substantial distance toward the storefront and finished close to it. Direction-aware, so it separates walking up to the shop from passing at constant distance.
- touching_display — a wrist inside a storefront display rack while still outside. Several entrance racks face the walkway within arm's reach, so a passer-by can handle merchandise without entering. Of these cues, this gave the clearest evidence of attention in this footage.
- **lingering** — sustained presence while genuinely slow. Implemented, but **fires zero times** on this footage; nobody in this corridor actually stops.

Speed thresholds were taken from the measured distribution rather than chosen: the "walking" bound sits near its 75th percentile, the "slowed" bound at the median.

Why facing the store was demoted

The initial approach treated a detectable face near the store as sufficient evidence of attention. It flagged **61% of all passers-by** as interested — implausible for corridor traffic.

The problem turned out to be camera geometry rather than the threshold itself. The camera looks outward from inside the shop, so people walking down the corridor naturally present a detectable face. The cue was effectively measuring the camera's viewpoint, not the person's attention. This led to demoting facing to corroboration only: it can be recorded on a track that already satisfies a movement-based cue, but never qualifies one alone.

The trade-off is that a genuine glance without a change in movement can now be missed. I considered that preferable to counting a large amount of normal corridor traffic as interested.

Excluding people already inside

Review of the annotated video showed a customer **seated inside the store** labelled `INTERESTED`. The track had spent 4.6 minutes deep indoors, but **1.74% of its frames** fell marginally outside through bounding-box jitter at the threshold — enough to satisfy `slowing`, after which the verdict applied to the whole track.

I added a passer-by gate to handle this: cues are evaluated only for tracks that spend a majority of their life genuinely outside, for a sustained period, with a margin absorbing jitter. Tracks that entered are exempt, since entering is itself conclusive evidence of interest. Total Interested fell from 25 to 16.

Entry, and a rule that never fired

The storefront is open-plan with no door, so three walkable openings were annotated as boundary segments. Entry initially required a sign flip across the fitted storefront axis **and** a segment intersection on the same frame pair. Task 1 then reported zero entries.

Instrumenting the test showed why: 6 tracks produced a sign flip, 7 intersected a segment, and **zero did both on the same frame pair**. Two independent geometric tests do not agree instantaneously when the global best-fit axis and the individual short openings are not perfectly aligned.

The fix was to move from an instantaneous to a **trajectory-level** test: start outside, end inside, and intersect an annotated opening at some point. A related latent defect surfaced alongside it — crossings were originally tested against the infinite line through each boundary, harmless when the storefront was believed to be one opening flanked by glass, but capable of firing on people nowhere near an opening once it was re-annotated as three short segments.

Entered vs passed by

Only one entry was detected across the 7.5-minute clip, so I checked whether tracker fragmentation might be hiding other entries: Of 80 tracks ending outside the store, only one has a plausible continuation beginning inside. If fragmentation were hiding entries, many such pairs would exist. The low conversion is consistent with what is visible in the corridor, subject to the entry-observability limitation described in the limitations section.

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4. Task 2 — Shelf Engagement

Why proximity alone was not enough

"Person near shelf" fails because of how people actually shop: they stand beside a fixture and reach into it, so their feet are outside the shelf polygon while their hand is inside.

I checked this against the footage rather than relying on the intuition alone. Across a sample of the interior clip, wrist-inside-region fires on 12.2% of person-observations against 10.8% for feet-inside-region — roughly 2.7× more often. Note precisely what that the measures: how often each signal is active, not which is more often correct, since no labelled engagement data exists. Combined with the physical argument, it was sufficient to make wrist position the primary evidence.

Wrist position also resolves the brief's "customer between two shelves" case with no special handling: the region containing the wrist is the shelf being engaged with. The hand decides, not the body.

One episode, not many

Each frame yields one of three values per person: a shelf name (wrist confirmed inside), none (wrist observed, in no shelf), or unknown (no wrist confidently detected). The third state is important. An occluded wrist is not evidence of disengagement, and collapsing it into "not engaging" would chop one

browse into several counted events — exactly the double counting the brief warns against. Unknown therefore sustains an open episode but can never start one.

Two thresholds turn this signal into events. **Minimum duration** rejects a noise floor visible in the data: three quarters of raw wrist-in-region runs last under half a second, which is a wrist clipping a polygon edge while walking past, while genuine engagement sits in the tail.

Gap tolerance defines when a return becomes a new event, and a sweep produced the key observation: the total moves only from 13 to 10 events across a 20× range, and is completely flat from 3 s to 10 s. The result suggests that genuine return visits in this footage are separated by more than ten seconds. That leaves a useful gap between a hand being withdrawn briefly during the same browse and the customer actually walking away and returning. Choosing the start of that plateau means the threshold can be off by roughly a factor of two without changing the reported count. The result is therefore not sitting on a narrow threshold.

Reporting uncertainty rather than hiding it

A separate, weaker signal was added: a person standing inside a shelf region for a sustained period without their wrist ever being confirmed there. People do not loiter inside a fixture by accident, but it is not proof of engagement — they may be queueing or waiting.

These are reported as **unsure** events in a separate column, never folded into the headline count. They surfaced six engagements the wrist signal missed entirely, including one at a shelf that otherwise recorded zero. Discarding them would under-report; counting them as confirmed would over-claim. Whether those six correspond to genuine engagement is not established — which is exactly why they are separated.

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5. Task 3 — Staff–Customer Interaction

This task exposed several important failure cases during development.

Why staff identification was hard

Staff wore an apron, which initially seemed straightforward. In practice, it was not: the store is full of warm wood shelving under warm lighting, so much of the scene occupies the same hue range as the uniform.

The initial approach approximated the torso as a horizontal band of the person's bounding box. Verified staff scored 56% red; a random sample of people scored 49% — essentially no separation. The problem was that the bounding box contained a lot of background, so the measurement was describing the room as much as the garment. (The random sample also contains the staff member, who is on screen constantly, so true separation is even weaker than the seven-point gap implies.)

I therefore switched to a torso region built from the shoulder and hip keypoints — a polygon containing garment and little else, using keypoints the pose model was already producing at no additional cost.

I chose colour over a learned appearance embedding because the useful visual cue was already known: the apron has a distinctive colour. An embedding would have required the same reference crops to calibrate, added a dependency, and produced a score much harder to defend line by line. The downside is that this approach is scene-specific and would need to be reworked for a different uniform colour.

Why frame-level classification was insufficient

Classifying each frame independently made the label flicker: a single unfavourable angle flips a person between STAFF and CUSTOMER mid-track. Beyond the visual noise, it makes downstream interaction logic incoherent — a session would fragment every time the apron went briefly out of view. Classification is therefore applied to the track, aggregating per-frame scores into one verdict.

Why median aggregation failed

With a calibrated threshold and track-level aggregation, the staff member was still visibly labelled CUSTOMER in the output video.

The cause was the aggregation statistic. The apron is clearly visible in only part of any track — the staff member turns, bends down to fit a shoe, and is partly occluded by the customer being served. The median is dragged below threshold by the frames in which the garment simply is not in view.

To quantify the effect, I used a manually reviewed reference set: the manually verified staff crops in data/persons_dense provided the reference for what the uniform actually looks like, and measuring its saturation and brightness -- not just its hue -- was what finally separated it from the population. (That ≈148 s is a manually reviewed reference estimate, not exhaustive labelling.)

I switched to an upper percentile. In practical terms, this asks whether the clearest frames from a track contain enough apron evidence, rather than requiring the apron to be visible for most of the track.

Contradictory candidates, and the constraint that resolved them

Percentile aggregation recovered coverage but introduced a new problem: **nine pairs of staff-labelled tracks overlapping in time**, which is impossible when one staff member exists.

Sweeping the parameters showed no setting achieving both adequate coverage and zero overlaps — settings with no contradictions recovered barely a quarter of the manually reviewed reference estimate. The colour signal alone cannot fully separate staff from customers in this scene.

I then added a scene-specific constraint: among overlapping staff candidates, keep the one with the greatest total apron evidence and demote the rest. The ranking statistic mattered — ranking by score alone let a very short, very high-scoring track evict a long, solidly-scoring one, costing most of the recovered coverage. Weighting score by observed duration instead lifted staff recognition from 32 s to 127 s against the ≈148 s manually reviewed reference estimate, with no remaining contradictions.

The zero-overlap result needs to be interpreted carefully: the constraint removes contradictions by construction, so the result is partly guaranteed by the mechanism. It confirms the constraint works, not that every label is correct.

The same constraint did not hold for the interior video

Review then surfaced a clearly apron-wearing person in the interior clip still labelled CUSTOMER. Several interior tracks had been demoted despite scoring almost 1.0 for twenty seconds or more — near-total apron coverage over substantial durations.

The constraint had been applied globally, but it encodes a fact about `entrance.mp4`, not about the store. The interior view includes the service counter, where one staff member can work while another serves the floor.

The fix was to gate the constraint on evidence, per video: do two tracks both score very highly at the same time? On entrance, none do. On interior, several do. The constraint is now applied only to entrance.

Interaction sessions

A staff member and customer are interacting when the gap between their bounding boxes (zero when the boxes overlap) is within a threshold expressed as a **multiple of the staff member's box height**, not a pixel radius. The ratio matters: this camera has strong perspective, and a person at the back of the shop is roughly a third the pixel height of one at the front, so a fixed radius would mean metres deep in the scene and centimetres near the lens. Box height is a per-frame scale proxy, holding the rule at an approximately constant physical distance. The threshold was set just above the dense cluster in the measured distribution of normalised gaps.

Proximity is converted to sessions by the same temporal logic used for shelves: sustained presence to start, tolerance for brief lapses (the staff member fetching a shoe box is one session, not two), and closure beyond that so a later return counts separately.

The single-staff assumption

The staff-classified entrance tracks show zero temporal overlaps, and several are directly adjacent — one ending on the frame the next begins. That pattern is consistent with one person repeatedly re-acquired under new IDs, so they are merged into one logical instance, giving **16 sessions / 1 staff instance = 16.0**.

This is an important limitation: non-overlap is consistent with one staff member but does not prove it. Two staff who never appear simultaneously would produce an identical pattern. The denominator of 1 is the single most consequential assumption in the reported results — if it is wrong, the average changes proportionally.

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6. Key Experiments and Engineering Decisions

Question	Observation	Decision
Which tracker fragments identities least?	BoT-SORT produced fewer tracks and fewer fragmentation-consistent pairs than ByteTrack on both clips.	Adopt BoT-SORT.
Does appearance-based ReID help re-acquire people after occlusion?	It made fragmentation **worse**, not better — likely weak features mis-associating similarly-dressed shoppers.	Leave ReID disabled; the brief makes it optional.
Is global motion compensation useful on a fixed camera?	Disabling it coincided with track counts roughly halving on both clips.	Disable GMC. **Attribution is confounded** — that run also changed the detector, and no single-variable re-run was performed.
Proximity or hand position for engagement?	Wrist-in-region fires on 12.2% of person-observations vs 10.8% for feet (full clip). An earlier 2.7× figure came from a 900-frame sample and did not hold over the whole video.	Wrist position as primary evidence; body presence as a separate, weaker signal.
Can apron colour be measured from the bounding box?	Verified staff 56% red vs random population 49% — no usable separation.	Build the torso from shoulder/hip keypoints instead.
How should the uniform be measured?	Scoring red pixels against only the SATURATED torso pixels returned ~1.0 for dark, desaturated clothing, because the ratio was computed over 2-15% of the torso. Hue alone separates poorly here: the warm-lit store puts a quarter of all torsos in the red arc. The uniform	Divide by the whole torso, and gate on saturation and value as well as hue. Separation improved from 2.5x to 11.4x.

Question	Observation	Decision
	separates on saturation and brightness (staff median S=93/V=72 vs population S=48/V=111).	
Why did entry detection report zero?	6 tracks flipped side, 7 crossed a segment, 0 did both on the same frame pair.	Replace the instantaneous test with a trajectory-level one.
How sensitive is the shelf metric to its gap threshold?	Total events move 13 → 10 across a 20× sweep, flat from 3 s to 10 s.	Set the threshold at the plateau start; the metric is not knife-edge.

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7. Failure-Driven Development

I found the failures below by watching the annotated outputs and tracing individual cases. Several produced numbers that looked plausible, so visual inspection was necessary instead of treating the final counts as proof that the logic was correct.

Failure	Root cause	Fix
Seated customer counted as an interested passer-by	Cues evaluated on any marginally-outside frame; box jitter qualified a track spending 98% of its life indoors	Passer-by gate requiring sustained, genuine presence outside
61% of all passers-by flagged as interested	Camera faces outward, so a detectable face is a property of the geometry, not of attention	`facing` demoted to corroboration; a movement cue is required
Apron detector could not separate staff from shoppers	Torso approximated as a box band, which is mostly warm-toned background	Torso built from shoulder/hip keypoints
Customers labelled STAFF; staff labelled CUSTOMER	Apron score divided red pixels by only the SATURATED torso pixels, so a dark desaturated garment scored ~1.0 off a handful of warm-tinted pixels (2-15% of the torso). The gate also keyed on hue, which barely separates in a warm-lit store.	Divide by the whole torso; gate on saturation and brightness as well as hue. Staff-vs-population separation improved from 2.5x to 11.4x.
Two "staff" on screen simultaneously	Colour alone cannot fully separate staff from customers	Scene constraint: keep the candidate with most total apron evidence
Interior staff demoted to customer	The constraint encodes a fact about the entrance clip, not the store; the interior has a service counter	Gate the constraint per video on concurrent high-confidence evidence
Zero entries detected	Two independent geometric tests required to coincide on one frame pair	Trajectory-level entry test
Boundary crossings could fire far from any opening	Infinite-line test retained after re-annotation into short segments	Proper segment intersection
Diagnostic clips took six minutes to render 45 seconds	Correct track IDs at t=300s require re-tracking from frame 1	Cache all observations once per video

Failure	Root cause	Fix
One gate could not serve both cameras	The same uniform photographs differently in the two views; a gate tuned on the entrance rejected the interior uniform entirely.	Per-view gates and thresholds, with the interior treated as presentational only since no metric consumes it.

8. Results

Task 1: 17 interested → 1 entered / 16 passed by
Task 2: 10 confirmed shelf-interest events + 6 unsure events
Task 3: 6 interaction sessions / 1 staff instance → 6.0 sessions per staff member

These numbers are the outputs of the pipeline, not accuracy measurements. No labelled behavioural ground truth exists for this footage, so no accuracy, precision or recall figure is claimed anywhere in this document. All values match the generated CSVs and the annotated videos, which replay the same logic rather than recomputing it.

Task 1 — Store Interest and Conversion (`entrance.mp4`)

Metric	Count
Total Interested	17
Interested Entered	1
Interested Passed By	16

Task 2 — Per-Shelf Customer Interest (`interior.mp4`)

Shelf	Interest events	Unsure
Shelf_A	3	2
Shelf_B	6	4
Shelf_C	0	0
Shelf_D	1	0
Total	**10**	**6**

The same analysis was also run on `entrance.mp4` across its four annotated storefront zones (7 sure, 3 unsure), reported separately. The brief assigns Task 2 to `interior.mp4`.

Task 3 — Staff–Customer Interaction (`entrance.mp4`)

Staff instance	Sessions
Staff 1	6
Average sessions per staff member	**6.0**

9. Limitations and Production Considerations

No labelled behavioural ground truth. Every calibration relies on measured distributions, sensitivity analysis, indirect scene constraints, and manual inspection. Those methods locate errors and bound plausibility; they cannot quantify correctness. Finding no further errors is not evidence that none remain.

Single fixed camera. Occlusion destroys evidence irrecoverably. A long occlusion spanning a genuine departure and return merges into one event, and there is no second viewpoint to resolve it.

Entry observability. Entries can only be detected where the person was tracked while still outside. The detector does not acquire people until they are reasonably large in frame, so anyone first detected once already inside cannot be counted — a floor on the conversion metric.

Task 1's entry rule has one positive example. It has been exercised against a single observed entry; that is insufficient to establish general reliability across varied entry styles.

Track fragmentation. Task 1 counts should be read as distinct observed appearances, not guaranteed unique persons.

Staff identification depends on a visible apron. Roughly 40 s of staff screen time is recognised on the entrance clip. Interactions occurring entirely within an unrecognised stretch are missed, and the method would not transfer to a different uniform colour without re-deriving the colour gate.

Task 3 depends heavily on the single-staff assumption, and the supporting test cannot distinguish one staff member from two who never appear together.

Camera-specific thresholds and geometry. Speeds and distances are in pixels, and all annotated regions are valid only for these two viewpoints. They require re-derivation, not re-tuning, for another installation. The interaction proximity rule is the exception, being scale-relative by construction.

If this were taken into production

The pipeline is offline batch by design: it caches a full tracking pass, and Task 1 classifies a track by its completed trajectory. A production deployment would need:

- **Streaming operation** with persistent track state, so restarts do not lose open episodes. The temporal logic is already an incremental state machine and would port directly; entry classification would need restructuring.
- **A labelled evaluation set** — the change that would most alter the character of the project, replacing plausibility arguments with genuine precision/recall figures and allowing thresholds to be optimised rather than argued.
- **Ground-plane calibration** so spatial rules are expressed in metres rather than pixels, making thresholds portable across cameras.
- **Monitoring and drift detection.** Layout changes invalidate annotated geometry and a uniform change invalidates the colour gate — both fail silently, with counts simply falling.
- **Model, config and threshold versioning**, since reported metrics are only comparable over time if the parameters that produced them are versioned alongside.
- **Camera health monitoring** for lens occlusion, repositioning and focus loss, all of which degrade metrics without raising errors.
- **Multi-camera identity** to link a customer's journey from storefront to shelf, which no single view can provide.

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10. Conclusion

The main engineering work was in translating three ambiguous behavioural requirements into rules that can be applied consistently to the footage and explained afterwards: what counts as interest, when one engagement becomes two, and what to do when the evidence is simply absent.

The temporal reasoning layer was built once and reused across two tasks with different subject matter but identical counting semantics. Thresholds were derived from measured distributions and sensitivity sweeps rather than intuition, and where a metric proved insensitive to a parameter, that was established rather than assumed.

Most of the substantive corrections came from watching the annotated output and finding that plausible numbers were wrong — a seated customer counted as a passer-by, an attention cue measuring camera geometry, an apron detector measuring the room, and a scene constraint applied where it did not hold. Each was diagnosed by instrumenting the specific assumption rather than adjusting thresholds until the output looked reasonable.

The results are reported together with the limitations that affect how they should be interpreted. There is no labelled ground truth in this footage, so no accuracy is claimed; the single-staff assumption underlying Task 3 is stated as an assumption; and the code paths never exercised are identified as such. The important point is to be clear about what this system can support on this footage, and what it cannot establish.